Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/25/2010
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	HZE Radiation: Modulation Of Genetic Effects By RNA Interference Of NHEJ		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	12/31/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2009 per J. Dardano/JSC (3/2009) NOTE: Received NCE to 06/01/2009 per PI (7/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bedford, Joel (Colorado State University) Liber, Howard (Colorado State University)		
Grant/Contract No.:	NNJ04HD83G		
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Task Description:	For extended missions in space, astronauts will be exposed to a radiation environment unlike any on the earth's surface. Galactic cosmic rays include a particularly worrisome component of energetic iron nuclei which are high atomic number, high energy (HZE) charged particles with linear energy transfer (LET) similar to alpha particles but with very different track structures. While the relative biological effectiveness (RBE) of a particular radiation roughly depends on the LET, the very different track structures for these two radiations may well modulate this RBE even if they have identical average LETs. Further, this "radiation quality" for the exposures actually relevant for astronauts will be modified by interactions with materials in the spacecraft, including their own bodies. Basic to virtually all relevant biological effects of ionizing radiation is the underlying damage produced in DNA, and the subsequent cellular processing of such damage. The damage can be qualitatively different for different kinds of radiations, and the genetics of the biological systems exposed can greatly affect damage processing and ultimate outcome—the biological effect of concern. The accurate repair of DNA double-strand breaks (DSB) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosome rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. It is important, therefore, to understand cellular responses to DNA damage and to elucidate the mechanisms and phenotypic consequences of DSB misrepair. Our general aim in this proposal is to examine connections between cellular radiosensitivity phenotypes relevant for carcinogenesis, namely chromosome aberration formation, mutagenesis and telomere fusions, and deficiencies in various components of the non-homologous end joining system, using RNA interference to systematically knockdown gene products known to be involved, either singly or in combination.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This project utilized the powerful technique of RNA interference (RNAi) to deplete levels of various components of the non-homologous end joining (NHEJ) double-strand break (DSB) repair system, then to study the effects on HZE radiation-induced chromosomal rearrangements, gene mutation and telomere dysfunction. The accurate repair of DNA double-strand breaks (DSBs) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosomal rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. We have focused on the consequent phenotypic effects of faulty repair by examining connections between cellular radiosensitivity phenotypes relevant for carcinogenesis after exposure to IR, and deficiencies in various components of the non-homologous end joining (NHEJ) system. In the past year, we produced deficiency of a protein called Apollo using RNAi in a human lymphoblastoid cell line. Exposure of cells exhibiting reduced protein expression to either gamma-rays or 1GeV/nu Fe particles demonstrated differential effects on telomere dysfunction and mutation frequency, as well as differential effects between radiation qualities. Apollo is a nuclease required for DNA interstrand cross-link repair, and in fact it appears to interact with FANCD2, also involved in that repair pathway. In addition, Apollo may be involved in a more generalized response to DNA damage, in that it may stimulate the activation of ATM, and it seems to interact with the MRN complex, which is a focal point for several modes of repair. The Apollo nuclease is also implicated in the protection of telomeres; it interacts with TRF2 in S-phase, and actually is an attractive candidate for the processing of telomere ends to generate 3’ single strand overhangs, a critical event and integral part of proper telomere formation. Knockdown of Apollo produced an increase in radiation mutagenesis, both for gamma-ray exposures and for 1 GeV/n Fe particles. The increases were of the same order as those caused by inhibition of DNA-PKcs, and furthermore there were no interactions, synergistic or antagonistic, of Apollo knockdown and DNA-PKcs inhibition. These results suggest that Apollo may be involved in the same pathway as DNA-PKcs, namely repair of DSB by NHEJ. We also examined telomere instability following siRNA knockdown of Apollo. We found telomere uncapping with knockdown of Apollo that was similar to that seen with inhibition of DNA-PKcs and similar between radiation types. These results suggest possible Apollo involvement in repair of IR-induced damage, as well as support a telomeric role for Apollo. Most recently, a collaborative effort with Dr. Sandy Chang at M.D. Anderson, TX in which SNMIB/Apollo knockout mouse embryo fibroblasts (MEFs) were generated, resulted in data implicating SNMIB/Apollo as a key 5’-3’ exonuclease required for generation of 3’ single-stranded overhangs at newly replicated leading-strand telomeres to protect them from engaging the non-homologous end joining pathway. A manuscript has just been accepted for publication in EMBO Journal. Our experiments have resulted in valuable information regarding CPR question 10.12, “What are the effects of inter-individual diversity?” We hypothesized that NHEJ repair components are major genetic determinants controlling individual susceptibility to radiation-induced cancer, as well as heritable effects of exposure. Therefore, naturally occurring deficiencies of these proteins within the population will put these individuals at increased risk. Our RNAi approach of experimentally mimicking highly relevant human deficiencies has shed light of this important issue. Our results support the relevance of deficiencies/hypomorphic mutations in NHEJ factors in that they can deregulate rejoining reactions sufficiently to increase incidence of oncogenic rearrangements (misrepair). Our work supports the concept that inherited or acquired mutations in NHEJ factors represent important risk factors for developing cancer, an especially relevant consideration following exposure to ionizing radiations (increased complexity the damage and its repair), and so also impact on CPR question 10.07, “What are the cancer risks in humans from spaceflight?” Our characterizations of telomere dysfunction, chromosome aberration formation and gene mutation serve as important and informative surrogate endpoints for predicting cancer risks.
Bibliography Type:	Description: (Last Updated: 12/10/2025)

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